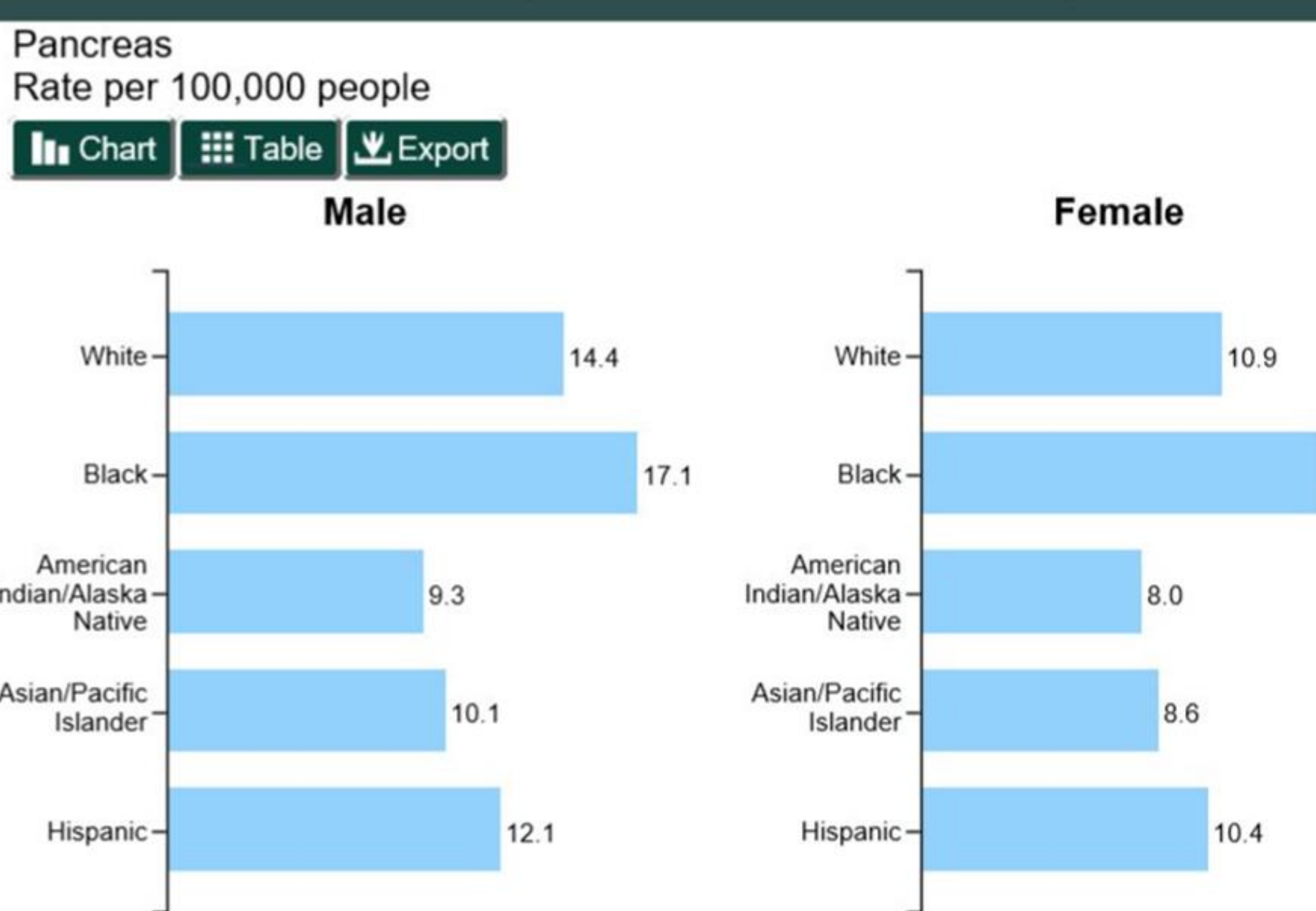


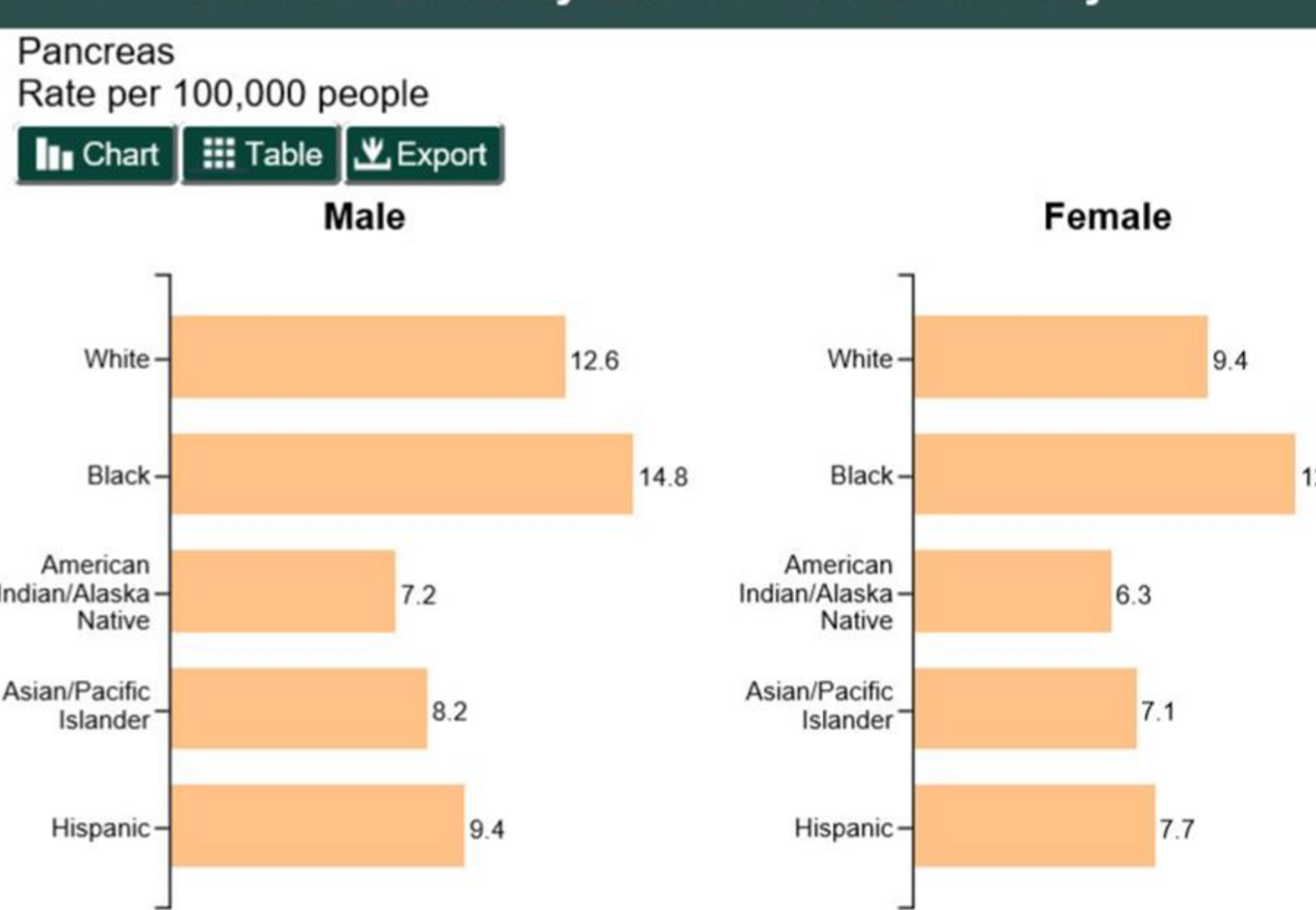
Introduction & Background

- Epidemiological investigations have found that long-term Type II diabetes mellitus (DMII) is associated with a 1.5-2 fold increase in the risk of pancreatic cancer (PaC).
- On the other hand, patients with new-onset diabetes diagnosed at an advanced age constitute a population in whom the PaC was the cause of DMII.
- Insulin resistance and associated hyperglycemia, hyperinsulinemia, and inflammation have been suggested to be the underlying mechanisms contributing to the development of diabetes-associated PaC.
- Signaling pathways that regulate the metabolic process also play important roles in cell proliferation and tumor growth. Use of the antidiabetic drug metformin has been associated with reduced risk of pancreatic cancer in diabetics and recognized as an antitumor agent.

Rate of New Cancers by Sex and Race/Ethnicity



Rate of Cancer Deaths by Sex and Race/Ethnicity

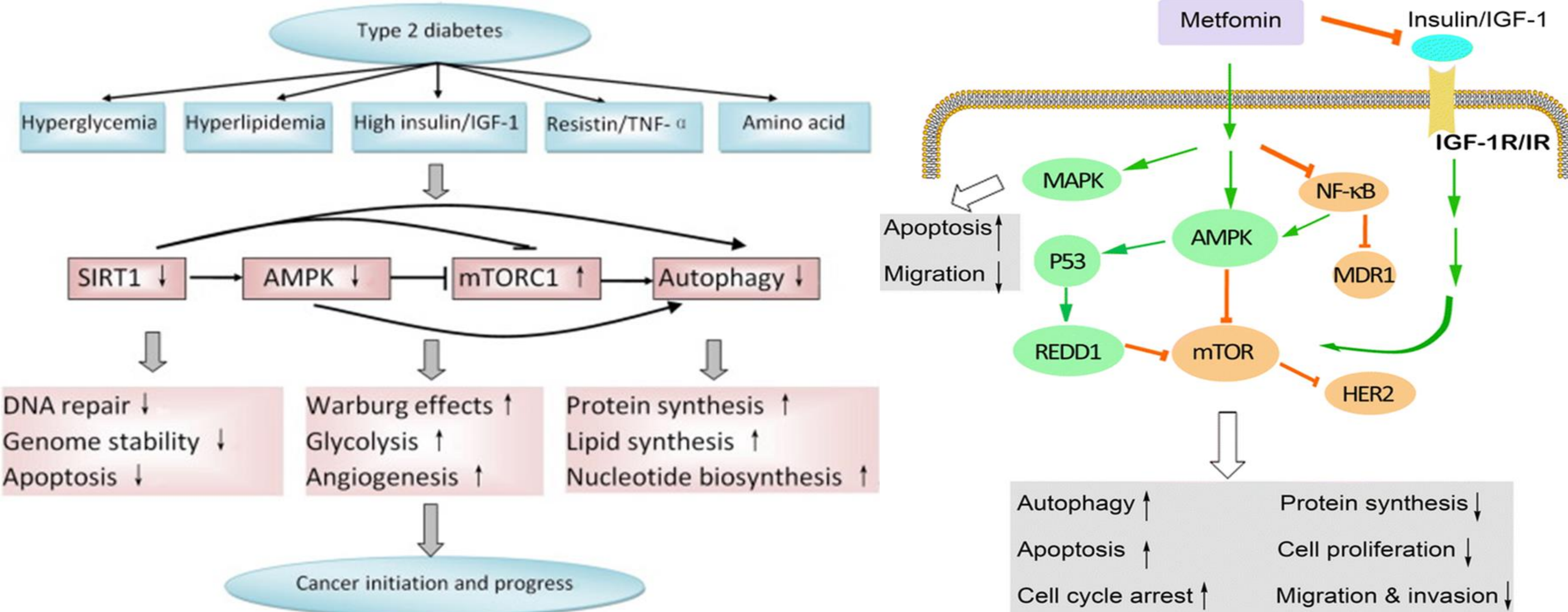


TOP

Age-Adjusted Diabetes Death Rates per 100,000 (2013)

	Non-Hispanic Black	Non-Hispanic White	Non-Hispanic Black/Non-Hispanic White Ratio
Male	45.1	23.1	2.0
Female	35.2	14.9	2.4
Total	39.5	18.6	2.1

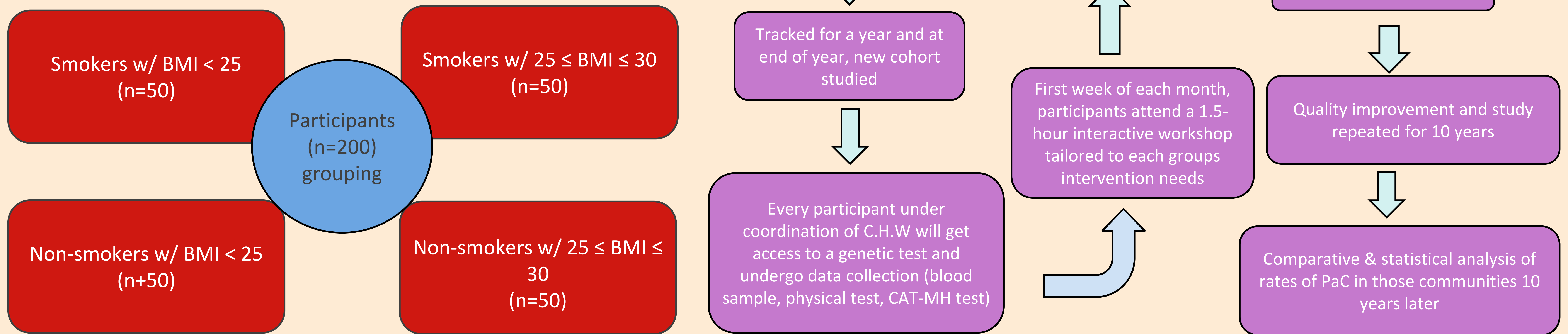
Source: CDC, 2014. National Vital Statistic Report. Vol. 64, Num 2 Table 17. http://www.cdc.gov/nchs/data/nvsr/nvsr64/nvsr64_02.pdf [PDF | 1.68MB]



Purpose & Research Methodology

By utilizing community health workers (CHW's) we hope to decrease rates of pancreatic cancer amongst African Americans in Chicago, by addressing its intersectionality with DMII and attacking it earlier on with the right educational and interventional methods, to weaponize participants with the information to make proper informed-decisions regarding their health conditions.

Disseminate CHW's in communities of southeast, southwest, south, and west areas of Chicago to recruit participants.



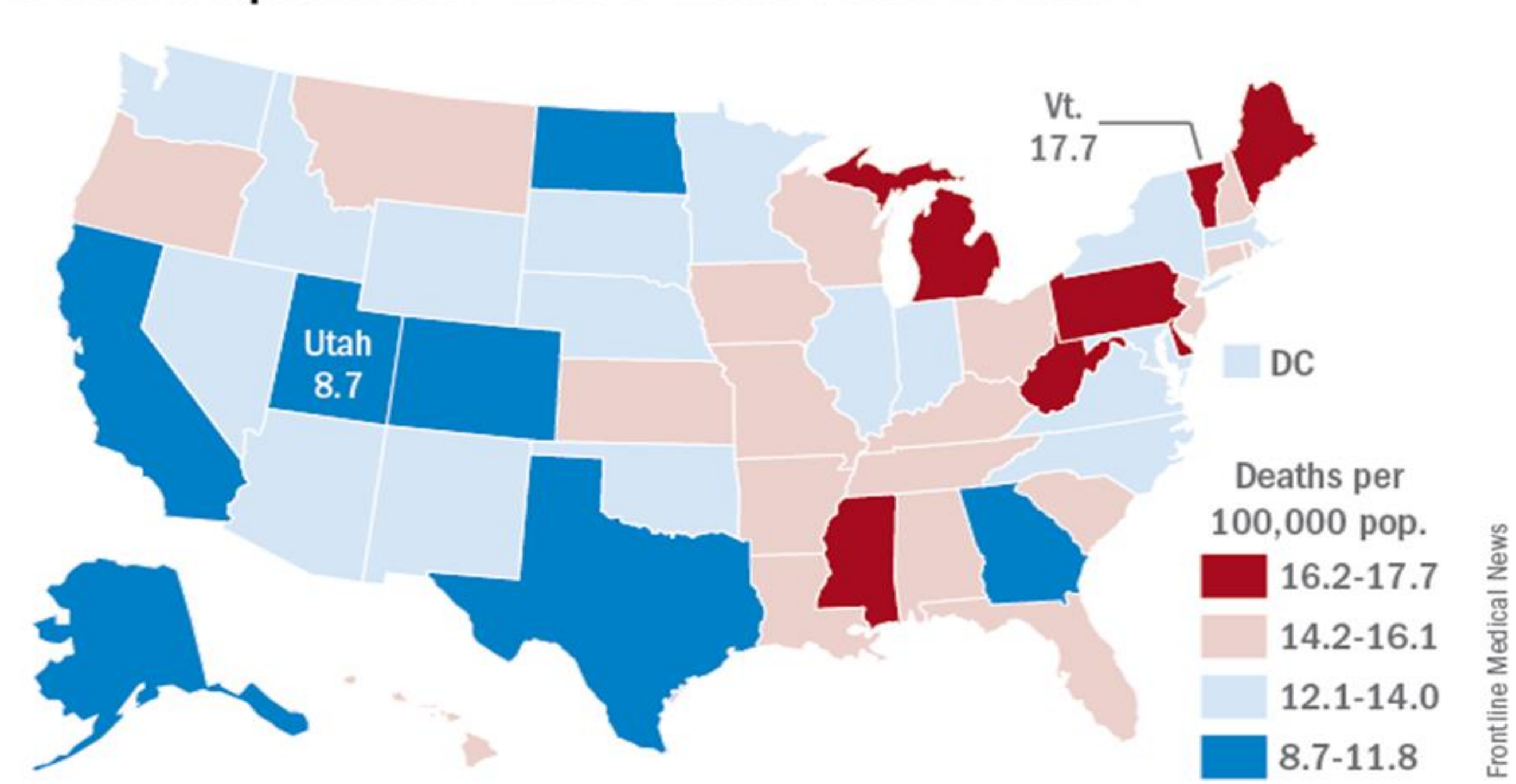
CHW Interventions

- Education on diabetes and its wide influences on health & illnesses, especially pancreatic cancer
- Deliver individual & group education on diabetes self-management and medication adherence
- Navigate participants through the complex health care system (how to set up appointments with PCP's, specialists, and other health professionals)
- Educate on smoking cessation and its impact on overall health
- Promote physical activity through exercise workshops
- Promote health eating habits and diet by teaching healthy cooking classes
- Help individuals get medical services if eligible. If not eligible due to insurance discrepancy, help them get access to the care by assisting in insurance securance
- Follow-Up and social capital & support system

Future Implications

- Sharing and usage of data to advance pancreatic cancer research and novel therapeutics
- In addition to blood samples and physical tests, the CAT-MH data can give valuable insight on the connection between mental health, diabetes, and pancreatic cancer.
- Blood samples of high-risk patients can be used to study biomarkers and molecular targeting techniques
- If the study is a success, can progress to the next stage in applying it to a national level and aimed at helping other populations at risk.

Estimated pancreatic cancer death rates for 2018



Note: Based on 2001-2015 mortality data from the National Center for Health Statistics. Source: American Cancer Society

Acknowledgements: Funding support is provided by the National Cancer Institute (grants U54CA202995, U54CA202997, and U54CA203000).

Note: The stated community intervention is not currently active, but is presented as a student proposal based on knowledge gained and background research conducted in the ChicagoCHEC Fellows Program. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health or ChicagoCHEC.